Two frameworks. One shared backbone.

Paper-inspired graph / temporal learning, extended with Temporal Residual Boosting (TRB).

1 PAPER FRAMEWORK - local implementation

Shared backbone for both methods. Graph, masks and transforms fit training data; future visits are excluded.

Available visits
Values + observation masks
Recorded time gaps
Public V=29; simulated V=52

→

Fixed graph + R-GCN
Two layers, node width 32
Latent code: 32 + 16
Observation-mask / input skip

→

MRTF
Past-only rolling windows
1 / 3 / 6 visits
Gated fusion to width 64

→

Temporal Transformer
Two layers, four heads
Past-only attention + gaps
Temporal embedding h_t:64

→

Factual path + heads
Noise / treatment-action path
Flare + severity mixture
Three Gaussian components

→

Paper output
Deterministic prediction
Validation Platt calibration
Flare probability + severity

2 PROPOSED FRAMEWORK - the additional stage

The neural architecture stays shared; the complete predictor gains explicit history inputs and a second fitting stage.

Reuse the Paper network
Same frozen neural weights
Paper margin / severity starting point
Temporal embedding h_t

→

Add observed history
Nine past-only groups per variable
Plus gap and h_t: 9V + 65 inputs
326 public / 533 main simulation

→

Temporal Residual Boosting
Classification with Paper base margin
Regression on severity residuals
Validation selects settings / strengths

→

Proposed output
sigmoid(ell_P + alpha * delta)
clip(s_P + beta * residual, 0, 4)
One proposed framework, two task heads

Why add the correction?

Compression can leave useful measurements and trends insufficiently expressed. TRB sees current values, missingness, running / recent history, changes, rates and observation age. It learns nonlinear corrections without replacing R-GCN, MRTF or the TCT. XGBoost is established; the contribution is this temporal integration.

How the methodology changes

Stage 1: fit the shared neural model with validation early stopping. Stage 2: freeze it and fit correction heads on training patients. Validation selects settings and correction strengths; test patients only score the final pair. Neural seed 73 and split seed 42 are frozen defaults. No future outcome is a prediction input.

How to interpret the evidence

Current scores and gains are generated in docs/MEASURED_RESULTS.md and its JSON snapshot. Read all metrics and bootstrap intervals, including unfavorable changes. The authors' original clinical cohort is unavailable. Local gains do not prove superiority on that cohort, universal improvement, identified treatment effects or clinical validity.

Paper source: Sharma & Kaur (2025), IEEE Access 13, 150010-150027. DOI: 10.1109/ACCESS.2025.3598276.

Publication: Sections III-IV and Algorithm 1. Local adaptations and mathematical details: docs/METHODOLOGY.md and CAUSAL_MODEL.md.

LECG is disabled in both active backbones. The two correction task heads are part of one Proposed framework.

This sheet contains architecture, not fixed performance numbers. See MEASURED_RESULTS.md for the latest verified comparison.

Academic research prototype. Temporal masking and causal terminology do not establish clinical causal identification.